

Download and Install Java - Windows Operating System

Java is a versatile programming language widely used for building applications. To start coding in Java, you first need to install the Java Development Kit (JDK) on your system. This article provides detailed steps for installing Java on Windows 7, 8, 10, 11, Linux Ubuntu, and macOS.

Here, we will discuss how to download and install Java on a Windows 64-bit Machine and **set up the environment** to run the first Java program on the command prompt. The process is the same for all Windows i.e. 7, 8, 10, and 11.

Step 1: Visit the official Oracle website <https://www.oracle.com/in/java/>. Click on the "**Download Java**" icon as shown in the below image.



Step 2: Now you can see the latest version is **JDK 23** and there are options for Linux, macOS, and Windows. Click on the **Windows** option and then click the **x64 Installer** option to download the .exe file for 64-bit Windows OS.

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Java 23, Java 21, and earlier versions available now

JDK 23 is the latest release of the Java SE Platform.

JDK 21 is the latest *Long-Term Support (LTS)* release of the Java SE Platform.

Earlier JDK versions are available below.

JDK 23 JDK 21 GraalVM for JDK 23 GraalVM for JDK 21

JDK Development Kit 23.0.1 downloads

JDK 23 binaries are free to use in production and free to redistribute, at no cost, under the [Oracle No-Fee Terms and Conditions \(NFTC\)](#).

JDK 23 will receive updates under these terms, until March 2025, when it will be superseded by JDK 24.

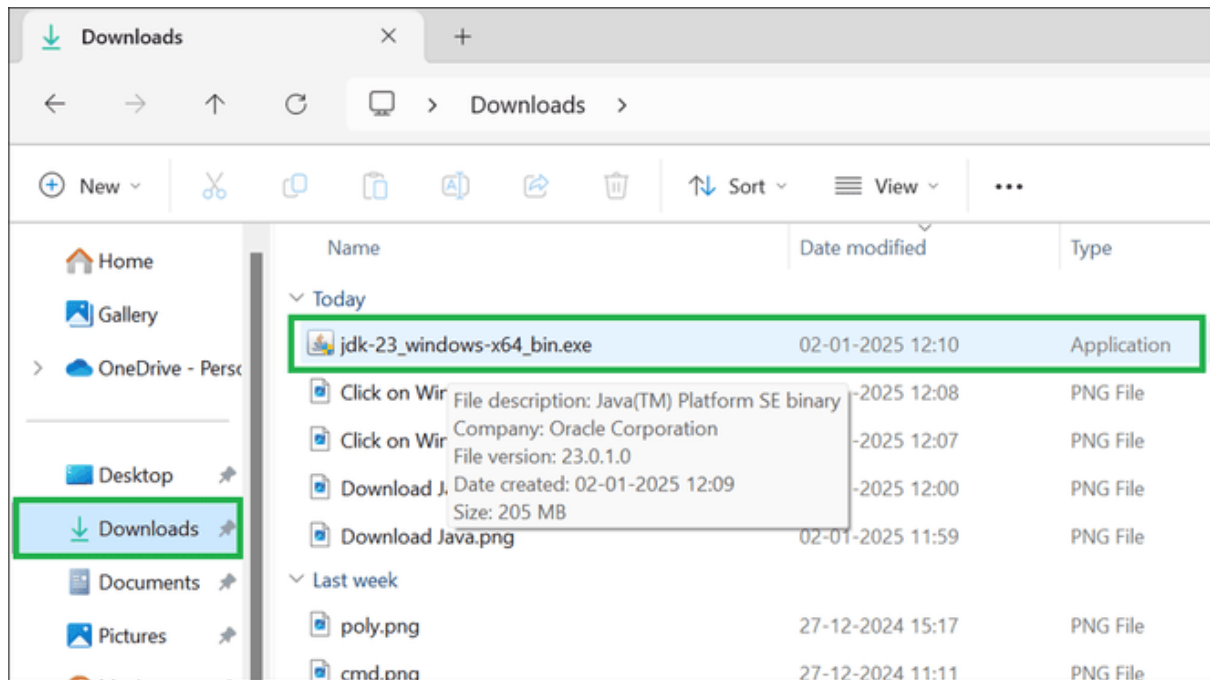
Linux macOS **Windows**

Product/file description	File size	Download
x64 Compressed Archive	228.70 MB	https://download.oracle.com/java/23/latest/jdk-23_windows-x64_bin.zip
x64 Installer	205.21 MB	https://download.oracle.com/java/23/latest/jdk-23_windows-x64_bin.exe
x64 MSI Installer	203.96 MB	https://download.oracle.com/java/23/latest/jdk-23_windows-x64_bin.msi

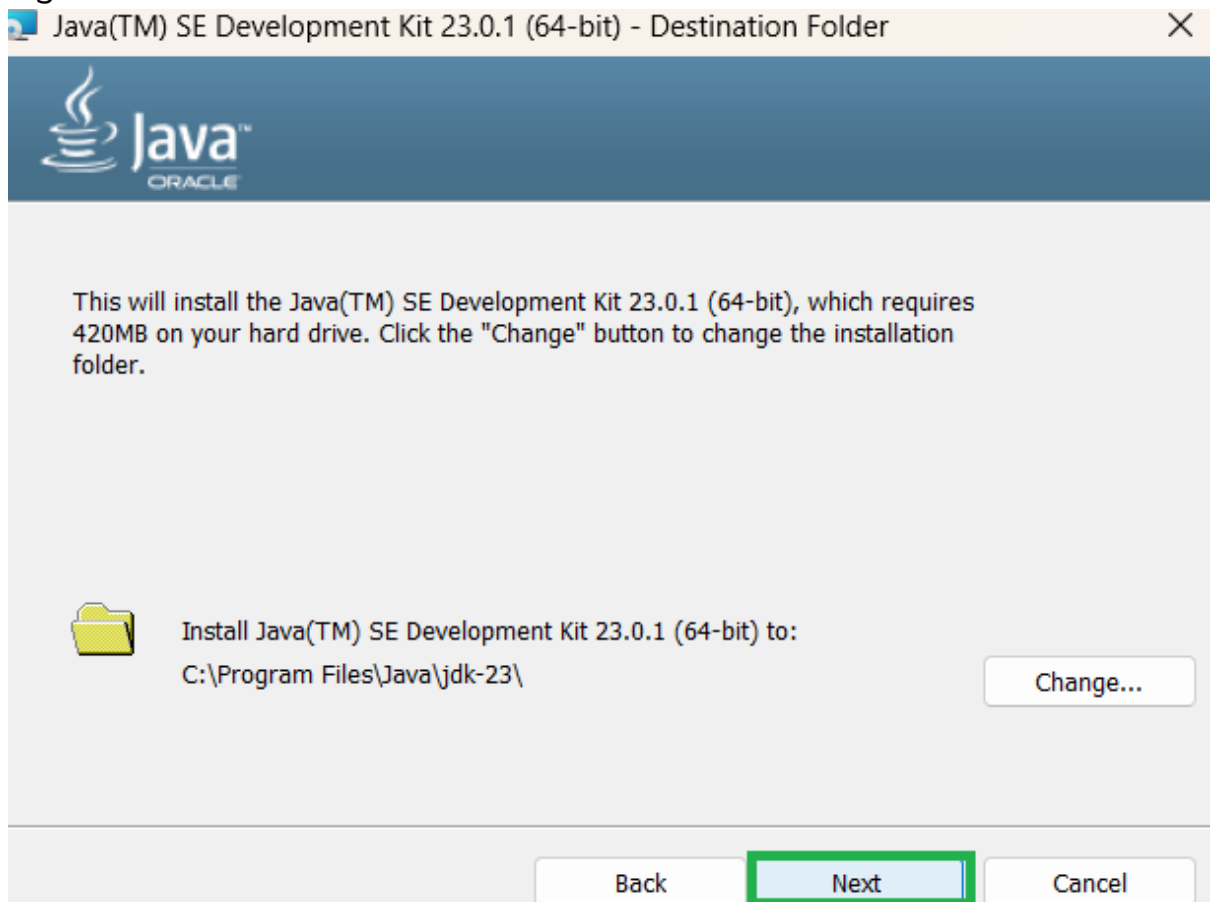
va/23/latest/jdk-23_windows-x64_bin.exe

You can simply use this **link** to download Java for Windows 64 Bit. After successfully downloading JDK 23, now we need to install the Java on our Windows System.

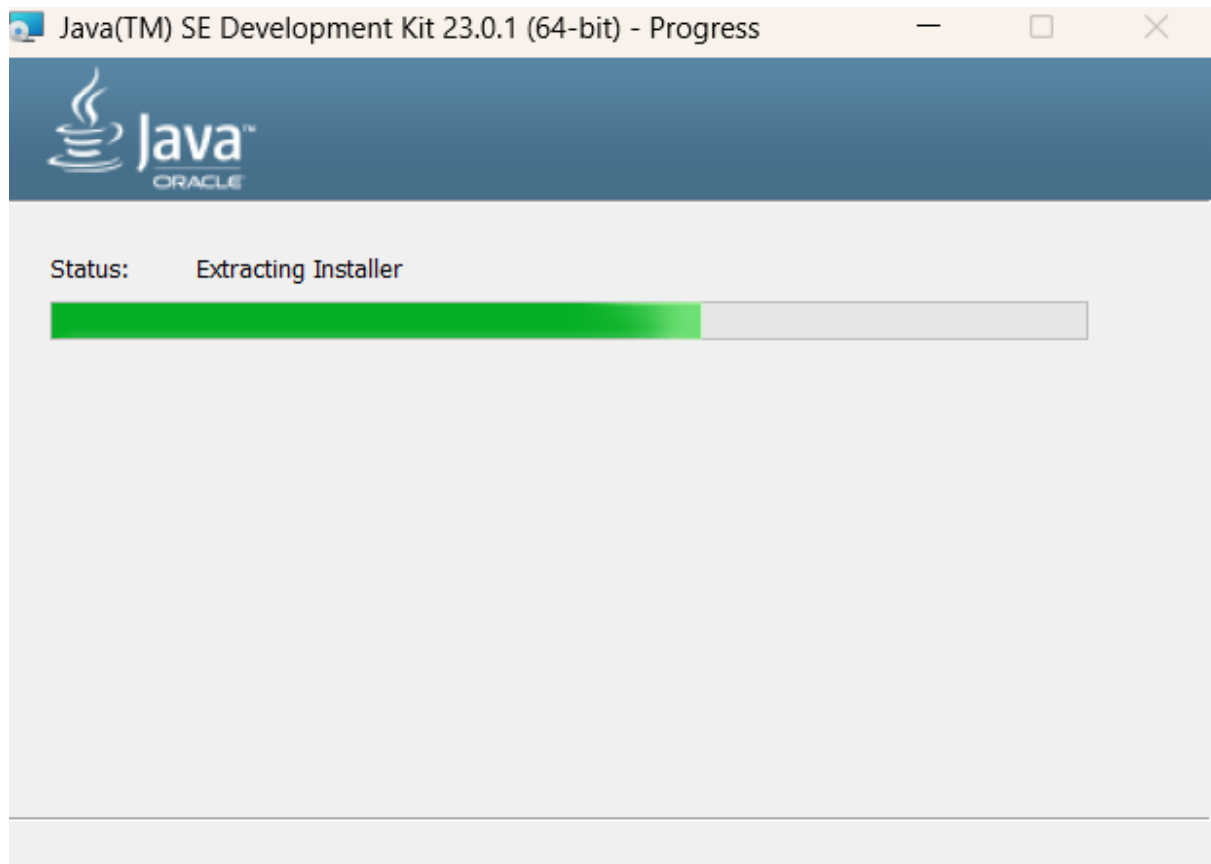
Step 3: Go to **Downloads** and double-click on that downloaded **jdk-23_windows-x64_bin.exe** file. So now **Java Installation Wizard** get open and then click **Next** button.



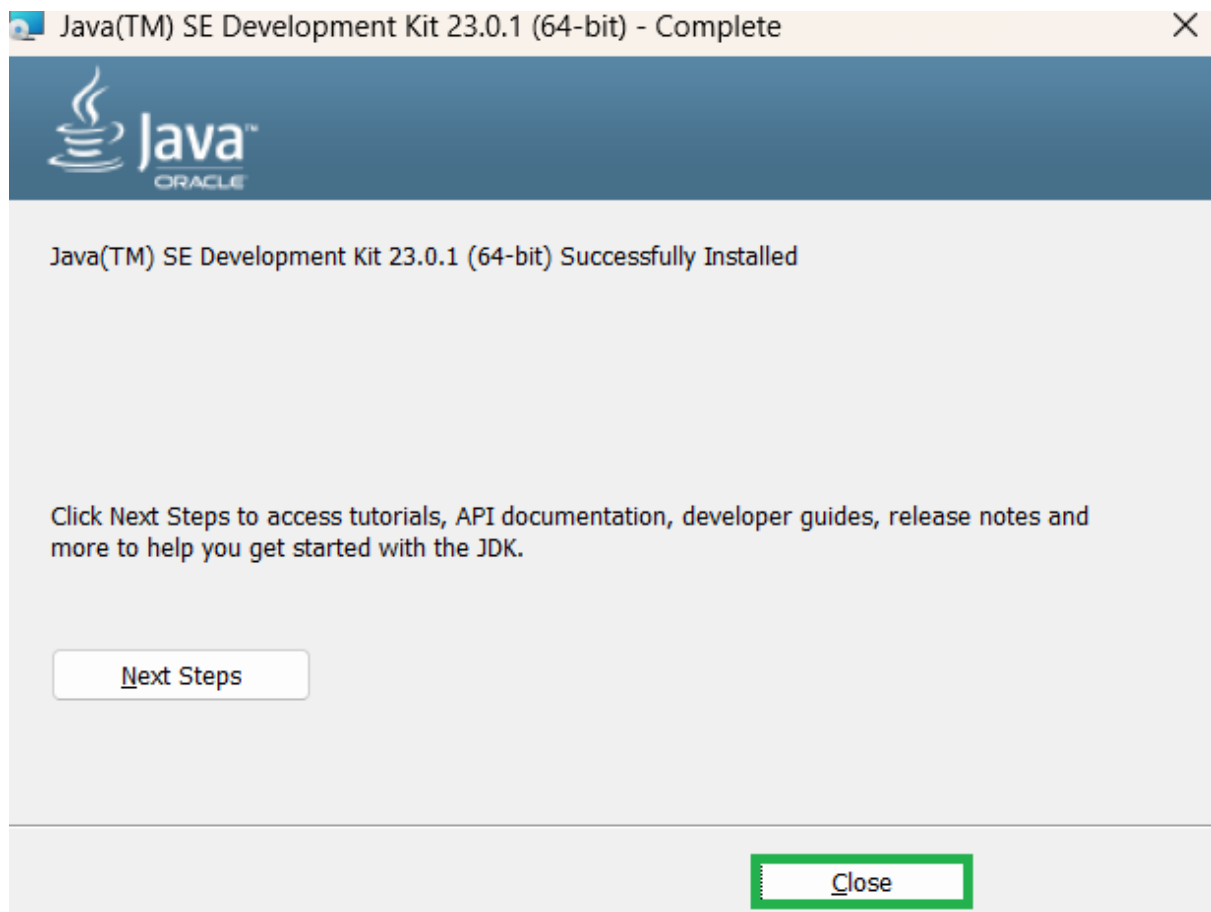
Again click on the **Next** button to install JDK.



The installation will begin as shown below:



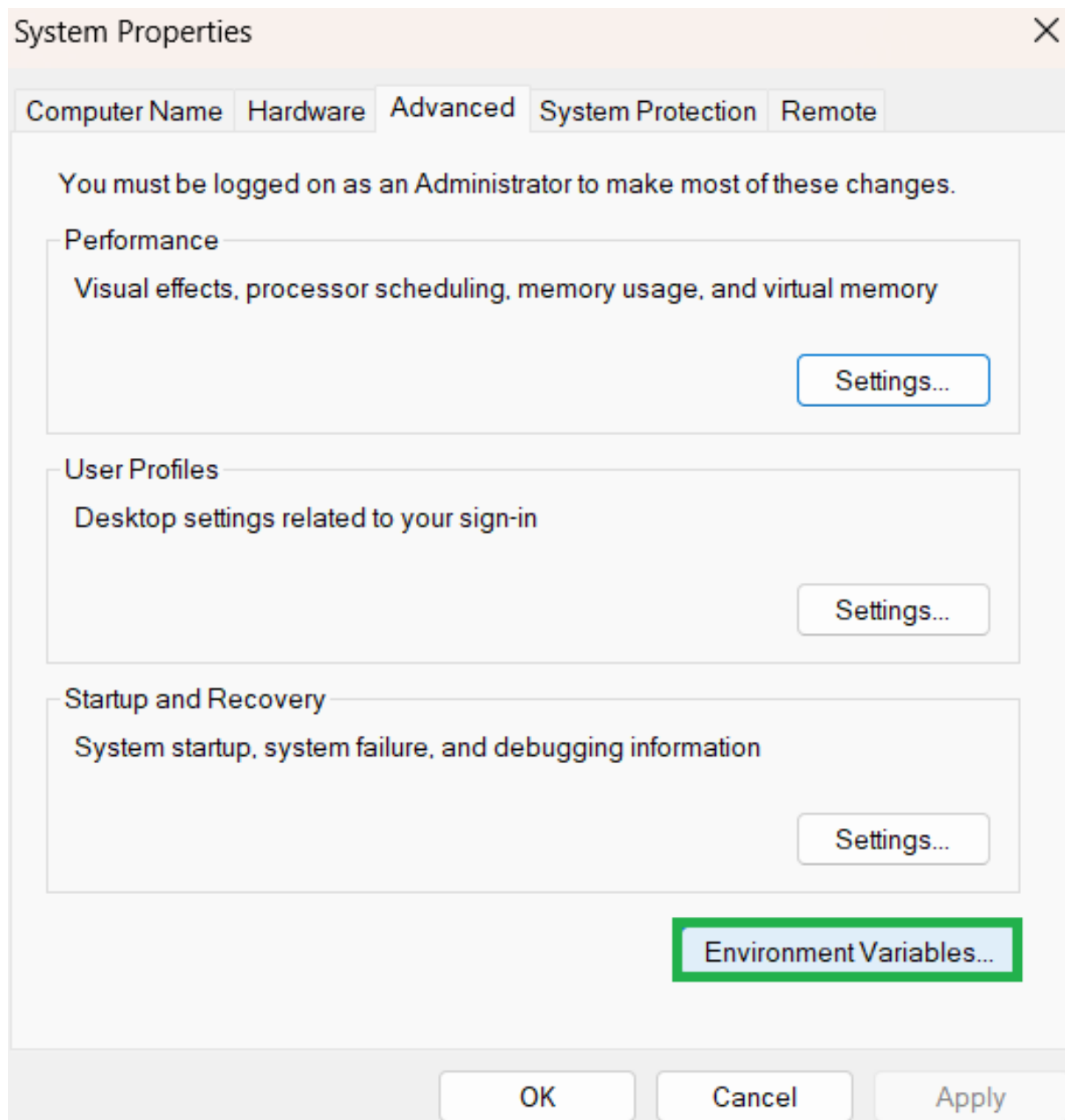
The Java installation is successfully completed as shown below. Now click on "**close**" to finish.



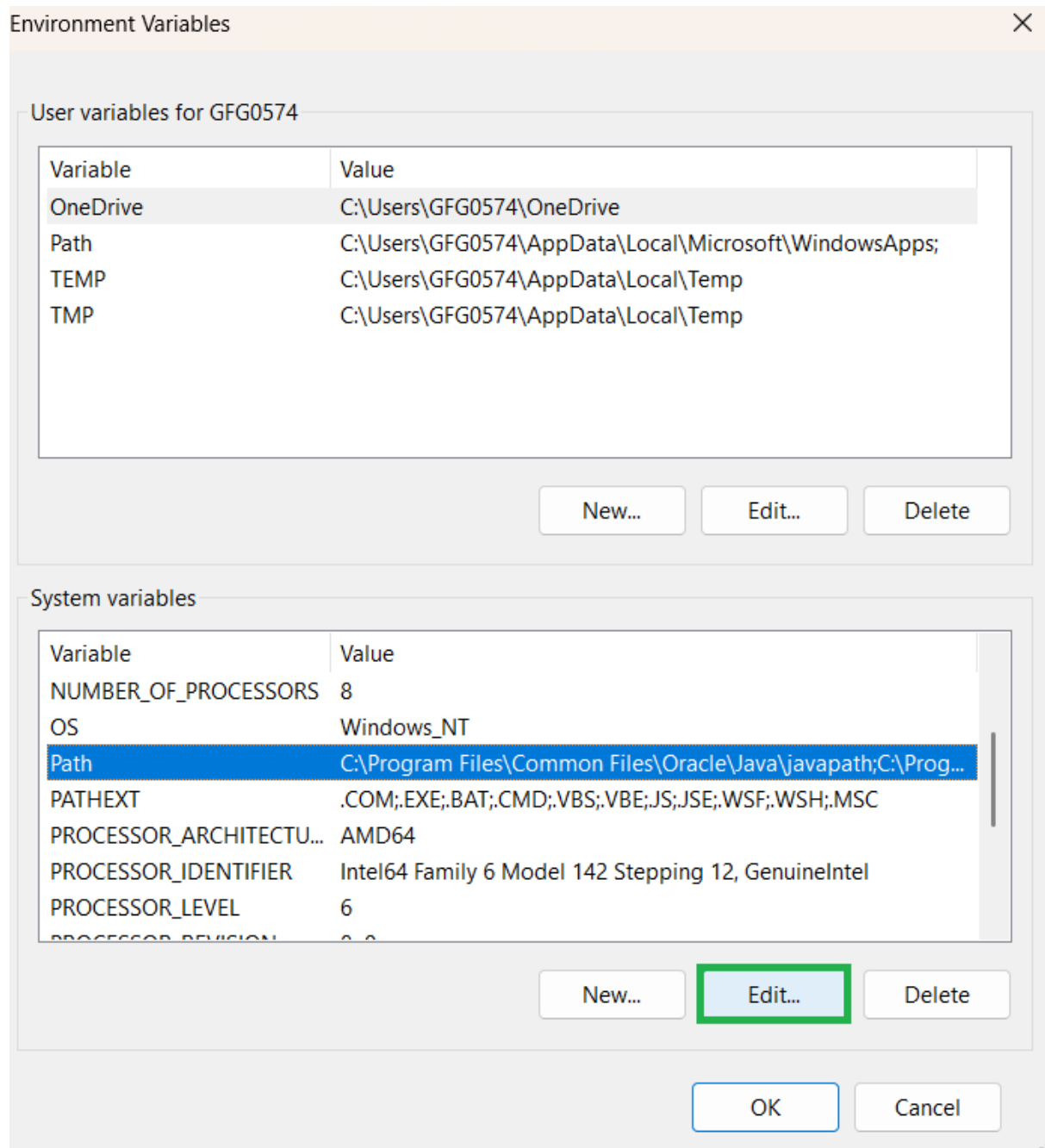
Step 4: Now we will set the environment variables for Windows OS. Open the **C drive**, go to **Program Files > Java > jdk-23 (or your installed version) > bin folder**. Now, copy the path and we will use this path when configuring the environment variables.

C:\Program Files\Java\jdk-23\bin

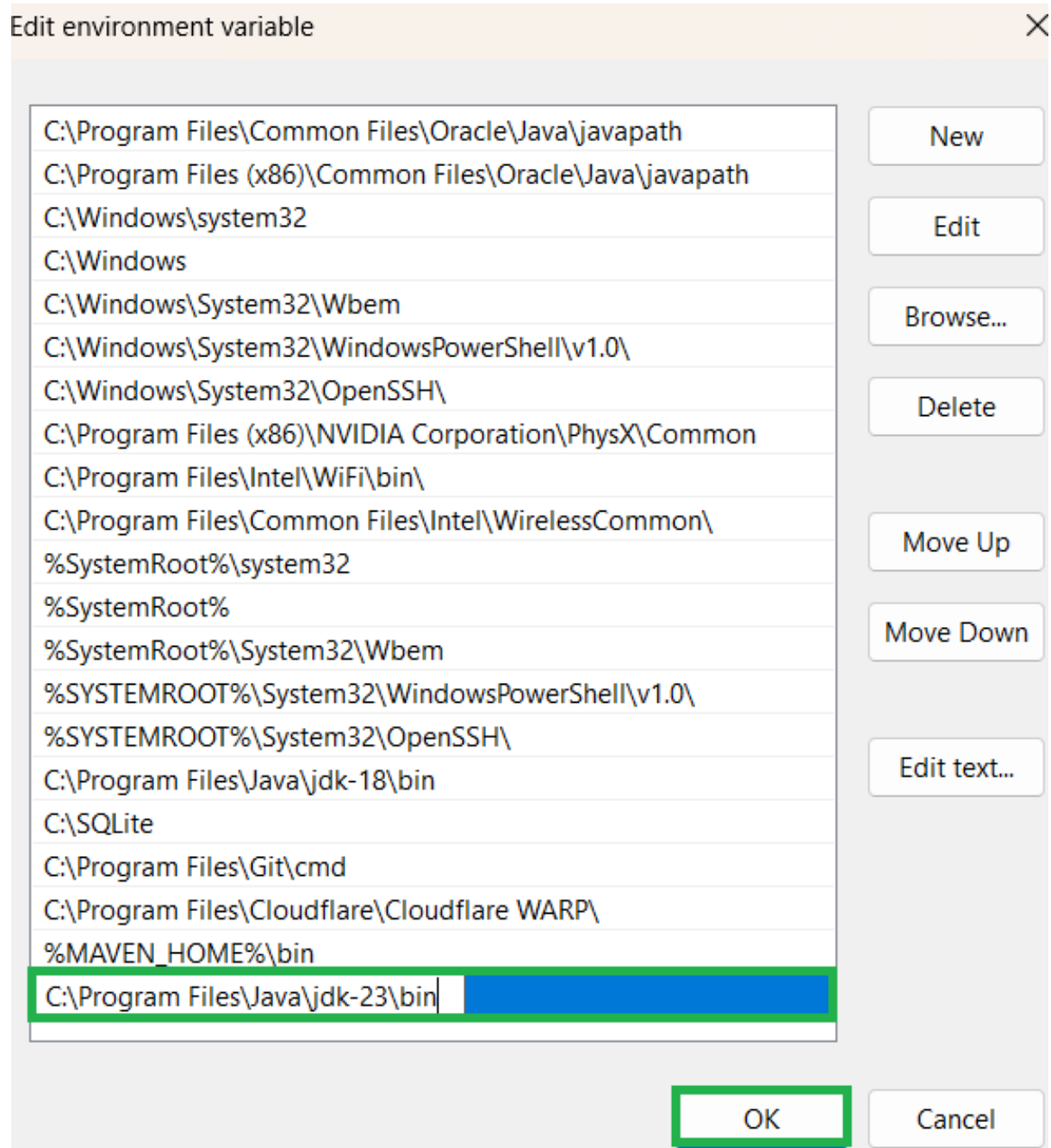
Search for environment variable on your system, then click on the Environment Variables button.



In the system variable section, select the path variable and then click the option Edit.

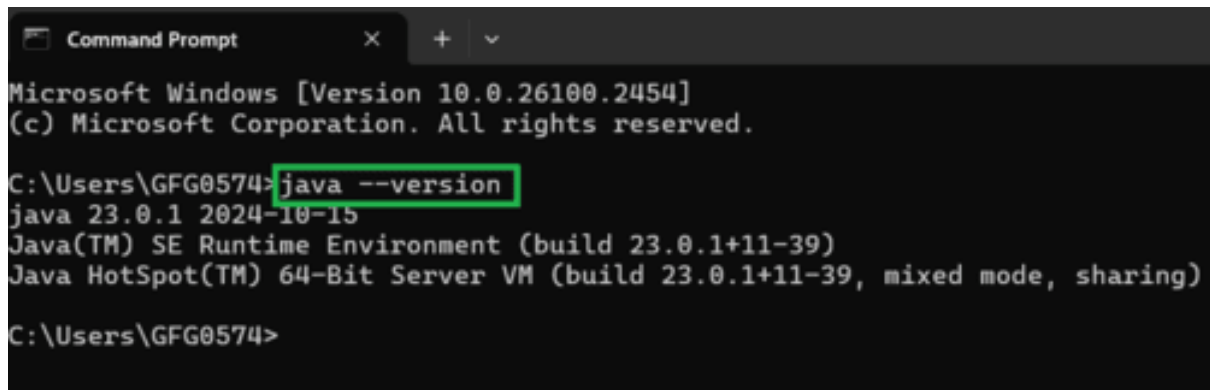


Paste the path that copied earlier, and **click OK** to save the changes.



Step 5: Verify the Installation. Open command prompt and type the below command **to verify the Java version that is installed in the system.**

`java --version`



```
Microsoft Windows [Version 10.0.26100.2454]
(c) Microsoft Corporation. All rights reserved.

C:\Users\GFG0574>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\GFG0574>
```

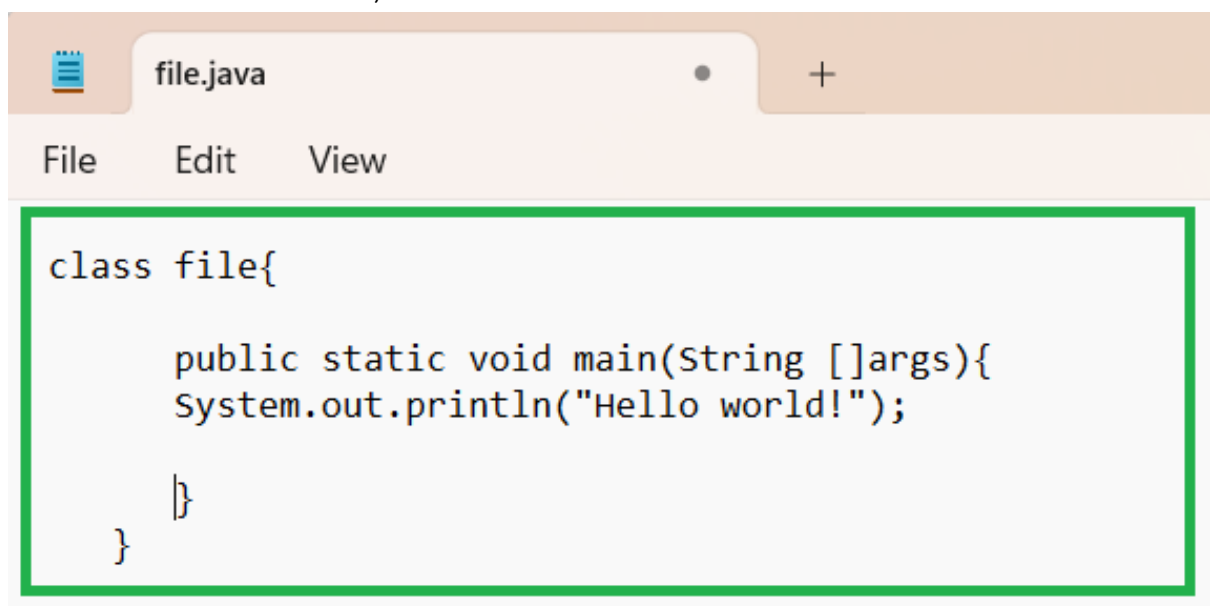
Now let us run our first Java program, to do this, follow the below steps:

Step 1: Open any **Drive**, and in the drive, create a **text file** and then change the file name with **.java** extension as shown below:

file.java

Step 2: Now, write a simple Java code in notepad.

- Right-click on the *file.java* and then select **open with > Notepad**.
- Inside the file, write the Java code.



```
class file{

    public static void main(String []args){
        System.out.println("Hello world!");
    }
}
```

Important Point: In Java, the class name should be equivalent to the file name. In this case, the file name is *file.java*, so the class name inside the file should be *file*.

Step 3: Now, go to the directory where your Java file is saved.

Open command prompt and navigate to the **D drive**, where the *file.java* is located. To do this, type "**dir**" as shown below:

```
Command Prompt
Microsoft Windows [Version 10.0.26100.2454]
(c) Microsoft Corporation. All rights reserved.

C:\Users\GFG0574>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\GFG0574>D:
D:\>dir|
```

Step 4: The **dir** command display all the contents of the directory.

```
Command Prompt
C:\Users\GFG0574>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\GFG0574>D:
D:\>dir
Volume in drive D is New Volume
Volume Serial Number is A0FD-A2D5

Directory of D:\

07/02/2024  03:40 PM    <DIR>          Desktop
05/05/2024  05:38 PM    <DIR>          Downloads
12/23/2023  03:32 PM    <DIR>          Desktop
12/07/2024  06:43 PM    <DIR>          Desktop
05/12/2023  02:57 PM    <DIR>          Desktop
12/10/2024  01:00 PM    <DIR>          Desktop
09/26/2024  05:29 AM    <DIR>          Desktop
11/09/2024  01:07 PM    <DIR>          Desktop
03/27/2024  05:01 PM    16,175,625,766 file.java
02/29/2024  09:30 PM    <DIR>          Desktop
04/15/2024  02:28 PM    <DIR>          Desktop
12/27/2023  06:57 PM    <DIR>          Desktop
01/19/2024  07:46 PM    <DIR>          Desktop
                2 File(s) 16,175,625,886 bytes
                11 Dir(s) 22,298,038,272 bytes free

D:\>|
```

Step 5: Compile the Java program. The **javac** command is used to compile the Java program.

Syntax:

javac filename.java

To run the program use the below command:

java filename

Here, in this case the filename is **file**.

```
C:\Users\GFG0574>D:

D:\>dir
Volume in drive D is New Volume
Volume Serial Number is A0FD-A2D5

Directory of D:\

07/02/2024  03:40 PM    <DIR>          .Trash-1000
05/05/2024  05:38 PM    <DIR>          Download
12/23/2023  03:32 PM    <DIR>          Desktop
12/07/2024  06:43 PM    <DIR>          Eclipse code
05/12/2023  02:57 PM    <DIR>          Engagement-Data
12/10/2024  01:00 PM             120 file.java
09/26/2024  05:29 AM    <DIR>          java
11/09/2024  01:07 PM    <DIR>          java-code
03/27/2024  05:01 PM  16,175,625,766 java.rar
02/29/2024  09:30 PM    <DIR>          papa.pdf
04/15/2024  02:28 PM    <DIR>          questions
12/27/2023  06:57 PM    <DIR>          ring
01/19/2024  07:46 PM    <DIR>          sweet memories
                2 File(s) 16,175,625,886 bytes
                11 Dir(s)  22,298,038,272 bytes free

D:\>javac file.java

D:\>java file
Hello world!

D:\>
```

We have successfully run our first Java program on Windows OS.